JEROME S. PARPA

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SUMMARY

Licensed Electronics Engineer and Electronics Technician with a strong foundation in electronic circuits, semiconductor devices, analog electronics, and embedded systems. Experienced in developing microcontroller-based projects, implementing control systems, and applying electronics principles to practical engineering problems. Demonstrates analytical thinking, troubleshooting, and technical problem-solving skills. Seeking an entry-level Semiconductor Engineer position to contribute to device testing, performance evaluation, test data analysis, and product quality assurance.

EDUCATION

Bachelor of Science in Electronics Engineering **SY 2022 - 2026**

University of Rizal System – Morong Campus

WORK EXPERIENCE

Engineering intern / Laboratory Intern

Panasonic Manufacturing Philippines Corporation | May–August 2026

- Executed performance evaluation and electrical safety testing on consumer appliances, ensuring full compliance with internal Panasonic quality standards and safety regulations.
- Calibrated and configured test setups to collect precision electrical measurements and data, eliminating measurement error and producing accurate validation reports for engineering leads.
- Conducted packaging reliability assessments, including standardized vibration and drop testing, to identify structural vulnerabilities and minimize potential transit damage.
- Streamlined daily laboratory testing operations and documentation workflows, accelerating product verification turnaround and supporting quality assurance standards.

TECHNICAL SKILLS

Electronics and Semiconductor Fundamentals: Electronic circuits, semiconductor devices, analog and digital electronics, soldering, circuit analysis, diode and transistor operation, component testing

Test and Measurement: Oscilloscopes, digital multimeters, function generators, electrical measurements, waveform analysis, component testing,

Programming and Embedded Systems: C/C++, Python, Arduino, ESP32 microcontrollers, Raspberry Pi, microcontroller interfacing, sensor integration, serial communication, and embedded system prototyping.

Industrial Automation and Control: Programmable Logic Controllers (PLC), ladder logic fundamentals, motor control, and basic control systems

Engineering Software: MATLAB, Proteus Design Suite, Arduino IDE, Visual Studio Code, CAD software, Microsoft Excel, and Microsoft Office

SELECTED ENGINEERING PROJECTS

CNN-Based Fermentation Monitoring & Optimization System for Cacao Beans

Applied convolutional neural networks (CNN) to automate cacao fermentation monitoring, improving process consistency and quality control.

IoT-Based Multi-Parameter Health Monitoring System for Bedridden Patients

Engineered an IoT prototype integrating sensor hardware and microcontrollers for real-time, remote patient health tracking.

PID-Controlled Multi-Function Autonomous Robot

Designed and programmed a multi-purpose autonomous robot using PID control loops for precise line-following, maze-solving, and sumo mechanics.

AWARDS AND RECOGNITION

2nd Place – Artificial Intelligence, 2026 WorldSkills Philippines – Rizal | *TESDA Rizal Provincial Office* (Mar 2026)

Best Developmental Research, University of Rizal System (2026)